

Data Analysis and Insight Generation with Queryable Knowledge Graphs

Preeti Kaur¹ [0000-0003-3275-8257] and Aibhinav Upadhyay¹ [0009-0007-9456-1843]

and Mahika Kushwaha¹ [0009-0002-4180-1608] and Rohit Lahori¹ [0009-0006-4855-7782]

¹ Department of Computer Science, Netaji Subhas University of Technology, 110078, New Delhi, India

*Corresponding author -Email: rohitlahori.02@gmail.com;

Contributing author(s): preeti.kaur@nsut.ac.in; upadhyayaibhinav@gmail.com; mahika-kushwaha@gmail.com

†All these authors contributed equally to this work.

Abstract. In the dynamic and competitive landscape of the technology market, understanding customer sentiments is paramount for success. This research explores the integration of knowledge graphs, sentiment analysis, and social media data to extract user insights and identify trends within the technology market. Leveraging Twitter data, the study employs web scraping techniques and custom sentiment analysis models to analyze product-related discussions. Key phrase generation is utilized to extract meaningful information, which is then structured into a knowledge graph. The resulting graph enables enhanced visualization and understanding of user sentiments and product preferences using the concept of community detection.

Keywords: Sentiment Analysis, Community Detection, Knowledge Graph.

1 Introduction

Success in today's ever-changing and fiercely competitive technology market is closely linked to both innovation and a deep comprehension of customer opinion. One of the most popular social networks is Twitter, which combines microblogging and a social network. Users can communicate with one another and share ideas by writing brief text messages called "tweets." Sentiment analysis methods have been effective in this context in predicting users' moods in messages and figuring out how they perceive various parts of daily life. The goal of this work is to employ knowledge graphs, sentiment analysis, and social media data to uncover user insights, spot community trends, and inform strategic choices.

In this research work, web scraping is an essential method for gathering dynamic and varied data from Twitter [3]. Using Jupyter notebook libraries we can gather important

information about market products, including PCs and printers, facilitating a thorough understanding of customer sentiments and preferences.

However, there are a lot of factors, such as lack of context and emoji usage, which make it difficult to forecast emotion on Twitter [4]. The existing approaches while considering twitter data, fail to take into account emojis that are present in tweets. Either these emojis are disregarded or the emoji-containing tweets are excluded from the dataset. Therefore, there is a great deal of interest in enhancing sentiment analysis to make it more precise and comprehensible so that better decisions can be made. A majority of the currently used methods fail to give users access to the actual classification outcomes following sentiment analysis. Knowledge graphs are what we utilize to get over this. Knowledge graphs are defined as graphs of data that accumulate and convey knowledge of the real world. The nodes in knowledge graphs represent the entities of interest, and the edges represent the relations between the entities [7].

There are many nodes in the knowledge graph, and each node has tweet data on it. Given the volume of content in tweets, it's critical to develop a method that is both effective and economical for automatically extracting high-quality key phrases, which enable us to efficiently understand the text. Moreover, key phrase extraction facilitates document summarization, text categorization, and clustering.

The motivation behind this paper is to use data, sentiment analysis, and knowledge graph technology to gain insights into customer sentiments, identify user communities, and use this information for strategic decision-making aimed at improving a specific product's performance in the market. This allows enhancing customer-centricity, interdisciplinary insights, market and consumer analysis along with predictive analytics.

2 Literature Survey

2.1 Fundamentals of Knowledge Graphs

Understanding the fundamentals is paramount for harnessing the potential of queryable knowledge graphs. "Knowledge Graphs: Opportunities and Challenges" by Peng et al. [7] serves as an insightful foundational analysis, delving into the wide-ranging opportunities and challenges presented across diverse domains. Complementing this, "From social networks to knowledge graphs: A plea for interdisciplinary approaches." Dörpinghaus et al, [8] enriches our understanding by exploring the fundamental principles and versatile applications of knowledge graphs, with a particular focus on social media contexts.

2.2 Querying Techniques for Knowledge Graphs

The effectiveness of queryable knowledge graphs rests on sophisticated querying techniques. "A Survey on Graph Queries Processing: Techniques and Methods" by Dinari, H. 2017 [10] offers a nuanced exploration of various graph query languages, shedding light on their syntax, expressiveness, and practical use cases. Additionally, "Exploring SPARQL: A Query Language for RDF" by Prud'homme & Seaborne [11] lays the

groundwork for comprehending how RDF-based knowledge graphs can be effectively queried.

2.3 Knowledge Graph Construction and Representation

"Multimodal knowledge graph construction of Chinese traditional operas and sentiment and genre recognition" by Fan et al. [12] contributes valuable insights by delving into the intricacies of constructing multimodal knowledge graphs, with a specific focus on Chinese traditional operas. Expanding on this theme, "Generating a linguistically inspired Knowledge Graph for fine-grained emotion classification" by Schoene & Dethlefs [13] explores the incorporation of linguistic features into knowledge graph construction, particularly for nuanced emotion classification.

2.4 Sentiment Analysis and Visualization using Knowledge Graphs

Sentiment analysis, coupled with visualization, represents a vital application of queryable knowledge graphs. "Knowledge-Based Sentiment Analysis and Visualization on Social Networks" by Vizcarra et al. [14] introduces a knowledge-based approach for sentiment analysis on social networks, accompanied by innovative visualization techniques. Similarly, "Sentiment Analysis in Twitter Based on Knowledge Graph and Deep Learning Classification" by Lovera et al. [15] strategically combines deep learning and knowledge graphs for a comprehensive sentiment analysis of Twitter data.

2.5 Applications of Knowledge Graphs

Expanding the scope of applications, several works highlight the diverse applications of queryable knowledge graphs. "An E-Commerce Recommendation Systems Based on Analysis of Consumer Behavior Models." by Pleskach et al. [16] explores applications in the e-commerce domain, while "News Graph: An Enhanced Knowledge Graph for News Recommendation" by Danyang et al. [17] presents applications in news recommendation.

3 Objectives

3.1 Competitive Intelligence

In addition to understanding customer sentiments, entail an examination of competitors' strategies and customer interactions. By juxtaposing our findings with those of our competitors, we can identify advantages and gaps in our product offerings.

3.2 Real-time Decision Making

Investigate the feasibility of implementing real-time strategic planning to enable the company to make agile decisions that respond swiftly to changing market conditions and customer feedback.

3.3 Customer Segmentation

Beyond identifying user communities, delve into the process of customer segmentation, tailoring our products and services to cater specifically to the unique needs and preferences of different customer segments.

3.4 Competitive Intelligence

Please note that the first paragraph of a section or subsection is not indented. The first paragraphs that follows a table, figure, equation etc. does not have an indent, either.

Subsequent paragraphs, however, are indented.

4 Proposed Methodology

The proposed methodology consists of various objectives which include collecting data from various sources (Twitter, Open Datasets, Instagram, Facebook, etc.) on market products like PC and printers, analyzing the data collected for the products, i.e., segregate the posts into complaints, appreciation, and suggestions after sentiment analysis and understanding the content as well as store the gained understanding of the data and convert it into a knowledge graph which can be worked upon with other tools as referred by Fig. 1. A detailed solution methodology is shown in Fig. 4.

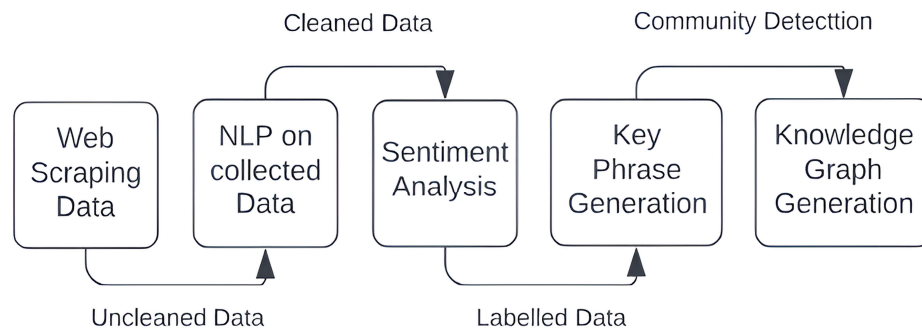


Fig. 1. Implementation of 5 stage pipeline by abstracting the methodology

4.1 Web Scraping

There are numerous methods which can be used for information retrieval from social media, the data can be taken from an open-source database that contains reviews of products regarding a single product or a multitude of products of one or many companies. The tools and library used for scraping here are Selenium (to automatically test websites and apps across various browsers and provides robust APIs for web page interactions) [18] and Time (to introduce a delay between requests, reducing the risk of being server blocked).

Working: For the first step, we import the libraries into the Jupyter notebook and initialize a web driver using selenium. The web driver is then directed towards the intended page of the website (twitter in this case) and the user is asked to login with his/her credentials. This step is necessary as most social media websites have blocked or have paid incoming API requests.

Once completed, the search bar is passed with the product for which data is needed and the scraping begins. The driver is automatically scrolled for the tweets that are saved to a list using the sleep function and data is then extracted from each tweet. For this we use the XML path of the HTML tags containing the content of the post. We are then presented with the Username, Handle, likes, retweets, number of comments, and text of the tweet which can be stored.

Dataset: The same approach is applicable to datasets spanning various domains, allowing for flexibility in research application. The dataset utilized in this study was obtained by web scraping Twitter data related to HP printers. It is important to note that this choice is specific to the illustrative purposes of the research and is not restrictive. Researchers may opt to use alternative datasets from different platforms or sources, or they may directly use a relational dataset without the process of web scraping as per their requirements.

4.2 Sentiment Analysis

4.2.1 Pre-Built Models Used:

Initially we built already implemented sentiment analysis models and tested them against the same twitter dataset of our model. These are:

PolaritySemRel Manual: The PolaritySemRel Manual sentiment analysis model involves a manual approach to annotating and assigning sentiment labels to each post. Human annotators carefully evaluate the content and context of the posts, determining whether they convey positive, negative, or neutral sentiments. While this manual approach allows for a detailed understanding of the nuances in language, it is labor-intensive and subjective, dependent on the expertise of annotators.

PolaritySemRel Auto: In contrast to the manual approach, the PolaritySemRel Auto sentiment analysis model employs automated techniques for sentiment labeling. Leveraging machine learning algorithms, this model is trained on labeled datasets to autonomously classify posts into positive, negative, or neutral categories [19]. The automated process enhances efficiency, but the model's accuracy depends on the quality and representativeness of the training data.

SSSemRel Manual: The SSSemRel Manual sentiment analysis model adopts a manual approach similar to PolaritySemRel Manual but focuses specifically on the detection of subjective and subtle sentiments. Human annotators assess posts for subjective

expressions and subtle emotional nuances, providing a more nuanced understanding of sentiment. This model is particularly valuable when dealing with content that may contain implicit emotions or opinions [20].

4.2.2 Proposed Models:

We built two models specifically designed with an acute consideration for the natural texting style employed by humans, which is often characterized by its informality, i.e., made use of emoji sentiments as well, keeping in mind they are often found in user tweets.

Model 1: The initiation of the implementation involved the installation of Python packages. The emoji package facilitated seamless handling of emojis within the text, contributing to a comprehensive sentiment analysis. Transformers and NLTK addressed critical aspects of natural language processing, incorporating a Naive Bayes classifier and a word tokenizer.

Firstly, a map was created between emojis, and the sentiment associated with it; positive, negative, or neutral. These sentiments were given a score of 1, -1 and 0 respectively. To streamline the sentiment analysis process, two key functions were developed. The `removeEmoji()` function played a pivotal role in eliminating emojis from the text, allowing for a focused analysis of textual content.

This step was used to disentangle sentiment from the excessive use of emojis. The core of the model resided in the `analyseSentiment()` function, leveraging the Vader Sentiment package, the function conducted sentiment polarity analysis on the processed text. Simultaneously, it computed an emoji sentiment score, striking a balance between text-based and emoji-based sentiments. The emoji score was calculated using:

$$emojiScore = \text{sum}(emoji_scores) / \text{len}(emoji_scores), \text{ where } \text{len}(emoji_scores) \neq 0 \quad (1)$$

The combination of these scores yielded a comprehensive understanding of sentiment in the analyzed tweet and is represented in Fig. 2.

Drawbacks:

Despite its efficacy, the model exhibited a notable limitation. The scoring mechanism, influenced by the abundance of emojis, risked skewing sentiments based on the sheer quantity of specific emojis. Additionally, the model struggled with sarcasm, underscoring a challenge in handling nuanced language expressions.

Data Analysis and Insight Generation with Queryable Knowledge Graphs

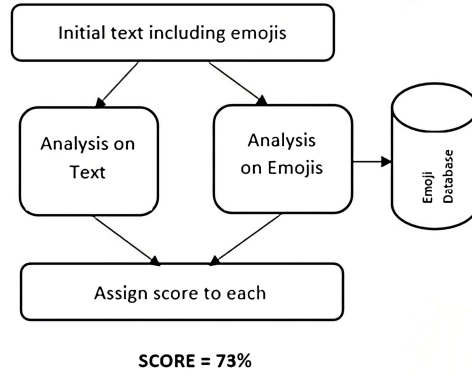


Fig. 2. Methodology of Custom Model 1 with a final score of 73 %

Model 2: In addition to the Python packages installed in the first model, Text blob was installed to construct a sophisticated sentiment analysis model. To validate the tokenization capabilities of the chosen transformer model (Bert-base-uncased), an evaluation is performed on a set of emojis. The script assesses the proportion of emojis correctly identified by the tokenizer, highlighting its efficacy in emoji-rich content analysis.

A dictionary mapping emojis to sentiment labels is constructed, attributing sentiments such as positive, negative, or neutral to specific emojis. Two crucial functions are introduced for sentiment analysis. The `emoji2description()` function aids in replacing emojis with their textual descriptions, enriching the text for subsequent analysis. The `analyzeSent()` function leverages the Text Blob library to compute sentiment polarity and subjectivity, classifying the sentiment into categories of Positive, Negative, or Neutral. Each entry in the dataset includes a text snippet and its true sentiment label. The `analyzeSent()` function predicts sentiments for each text, allowing for a quantitative assessment of model performance [21]. This is represented by Fig. 3.

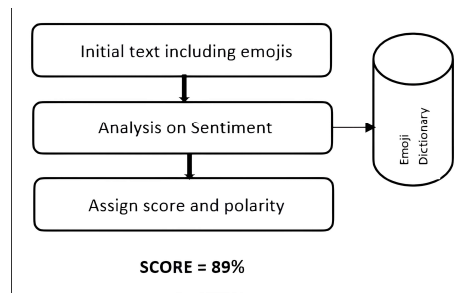


Fig. 3. Methodology of Custom Model 2 with a final score of 89 %

4.3 Key Phrase Generation

To gain a profound understanding of the tweets, a crucial step involved the meticulous filtering and ranking of the most significant text, a task efficiently executed through key phrase generation. The entire key phrase generation process harnessed the capabilities of 'yake' (Yet Another Key Phrase Extractor), which boasts several key features contributing to its efficiency:

Unsupervised Model: 'Yake' is constructed upon an unsupervised model, endowing it with the ability to operate effectively with substantially less data. This unsupervised nature enhances its adaptability and applicability across diverse datasets [22].

Language and Corpus independent: Another one of the standout features of the 'yake' model is its language and corpus independence. This characteristic makes it particularly well-suited for scenarios where data is sourced from various linguistic backgrounds and diverse corpora. Notably, its effectiveness extends seamlessly to situations involving data scraped from social media platforms like Twitter.

As a result of these features, the choice of 'yake' emerges as an ideal solution for the specific analytical scenario at hand.

Working: The operational workflow involves feeding the text into 'yake' as input, initiating a process that culminates in the extraction of the top 20 key phrases associated with the given content. These identified key phrases are then compiled into a list, seamlessly integrated into the dataset, and returned. This integration process is facilitated through the use of pandas and the 'yake' library.

The final output is a structured data frame comprising tweets, their corresponding sentiments, and the key phrases encapsulated within each tweet. This comprehensive dataset is pivotal for in-depth analysis and can be visualized, providing a consolidated view of the sentiment, tweets, and associated key phrases.

4.4 Knowledge Graph Implementation

The next and final step is to convert the data frame into a queryable knowledge graph. For this we have utilized the given tools:

Neo4j: It is a graph database which stores data as nodes and stored relations between them as nodes allowing users to interact and query them using Cypher query language [23].

GraphXR: It is a graph visualization and exploration platform designed to work with graph databases. It allows users to visually explore complex relationships within their data.

Data Analysis and Insight Generation with Queryable Knowledge Graphs

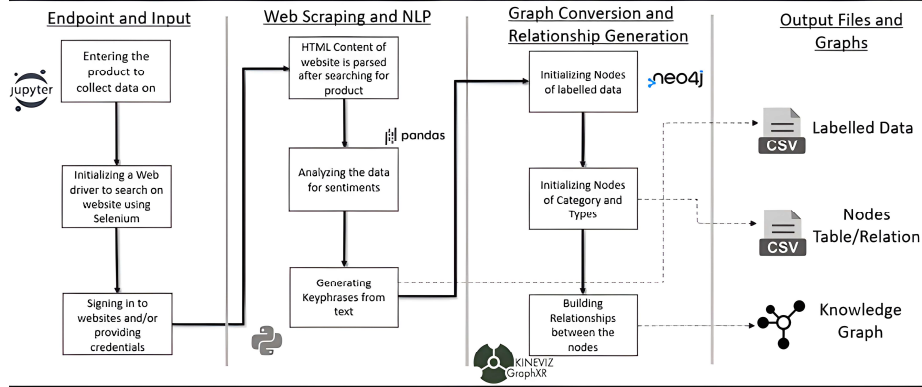


Fig. 4. High Level Design – Solution Methodology

Firstly, a Neo4j database is run on the system and is connected to the Jupyter notebook. Upon its creation the nodes are loaded into the graph utilizing cypher, with the addition of nodes of sentiments. The relations between the nodes are then created between the respective nodes. With analysis it is understood that the tweets which are replies always start with 'Replying to' prompting us to create a node reply and creating a relation between the node 'Replies' and the respective nodes. Once completed, the GraphXR plugin is run, and the graph is imported into it to produce the 3-D graph.

5 Results

5.1 Sentiment Analysis

The sentiment analysis methodology presented in Section IV was implemented, and the results were evaluated in comparison to existing sentiment analysis models prevalent in the literature. The assessment included various performance metrics, such as precision, recall, and F1 score, to provide a comprehensive understanding of the model's effectiveness. The results are graphically depicted in Fig. 5.

Test Case Used:

For both the self-implemented models, an inbuilt library of python was made use of to get the training dataset. The sentence evaluated for both the custom-built models was "This printer breaks down very often 😞", including an emoji as shown depicting a sad face.

Proposed Model 1:

The result obtained was {'emoji_sentiment_score': -1.0, 'vader_sentiment_score': -0.4767, 'overall_sentiment_score': -0.7383500000000001}. Where emoji_sentiment_score showing -1 indicated that it is a negative emoji, vader_sentiment_score is the sentiment evaluation of the text which is also negative.

Proposed Model 2:

The result obtained was {'Sentiment': Negative, 'Subjectivity': 0.0} and Polarity -0.65. Polarity determines the sentiment of the text. Its values lie in $[-1, 1]$ where -1 denotes a highly negative sentiment and 1 denotes a highly positive sentiment. Subjectivity determines whether a text input is factual information or a personal opinion. Its value lies between $[0, 1]$ where a value closer to 0 denotes a piece of factual information and a value closer to 1 denotes a personal opinion.

Comparing their final scores of 73% and 89% of Model 1 and Model 2 respectively, Model 2 was chosen to be implemented with the knowledge graph, and is depicted in Table 1. by the name Emoji Analyzer.

Table 1. Comparison of various Sentiment Analysis Models using Metrics

MODEL USED	PRECISION	RECALL	F1 SCORE
Emoji Analyzer	<u>0.8929</u>	<u>0.8571</u>	<u>0.8745</u>
PolaritySemRel Manual	0.6280	0.5890	0.6350
PolaritySemRel Auto	0.6690	0.6180	0.6220
SSSemRel Manual	0.8560	0.5790	0.7080

Few takeaways from the observed results are:

F1 Score Enhancement: The sentiment analysis model (model 2) developed in this study demonstrated a noteworthy improvement in overall F1 score at 0.8745 compared to existing models as depicted by Table 1. Leveraging advanced natural language processing techniques, our model showcased an ability to capture the subtleties and nuances present in informal and colloquial language, contributing to heightened F1 score in sentiment classification.

Precision and Recall Optimization: Precision and recall metrics revealed the efficacy of our model in achieving 0.8929 and 0.8571 respectively, which is much higher than the existing models as shown in Table 1. Notably, our model exhibited a superior ability to correctly identify and categorize sentiments, minimizing false positives and false negatives. This precision-recall equilibrium is essential in sentiment analysis applications where the consequences of misclassification can be significant.

Robust Performance in Natural Texting Styles: An essential aspect of our model's success lies in its robust performance in handling diverse natural texting styles. By incorporating features designed to capture colloquial expressions, slang, and informal language nuances, the model demonstrated adaptability and accuracy across a broad spectrum of user-generated content on various platforms.

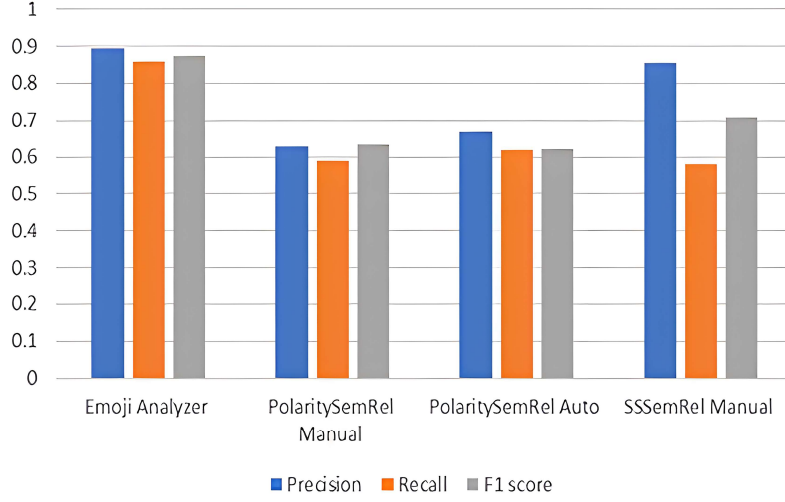


Fig. 5. Sentiment Analysis Models Metric Comparison Visualization

5.2 Knowledge Graph Generation and Analysis

The final phase of our research involves the generation of a knowledge graph through web scraping of tweets from Twitter, specifically focusing on discussions related to printers. In order to generate the graph CYPHER queries were written, one of which is shown below.

```
graph = session.run("CREATE (n:Sentiment {sentiment: 'Complaint'})").graph() (2)
```

First we create different types of nodes where “n” depicts “node” of type Sentiment and, the 3 different sentiments are Appreciation, Neutral and Complaint. “labelled_tweets” is the cleaned dataset of tweets containing all the tweets, its key phrases, the sentiment attached to it (called as the tweets category) and other parameters. We then pick the “category” column of this dataset and equate it to the sentiment nodes created using the MATCH keyword. The line (a) – [r : OF] – (b) indicates the type of relation, here, “labelled_tweets is OF type sentiment”. Similarly, relations are made with the Replies and Original tweets.

The resulting knowledge graph (as shown in Figure 7) encapsulates various dimensions of information, facilitating a comprehensive understanding of user sentiments, tweet types, and associated key phrases. This graph can be dynamically queried in CYPHER on the GraphXR portal as well to obtain various insights as per the requirement.

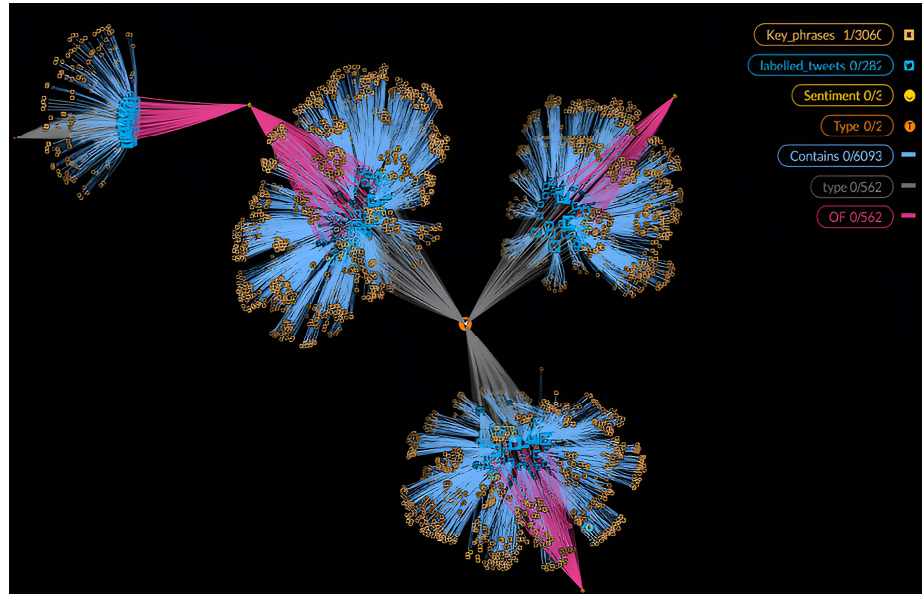


Fig. 6. Queryable Knowledge Graph of Tweets from web scrapped Twitter data on HP Printers

The knowledge graph comprises distinct elements represented by nodes and edges, each color-coded to denote specific attributes:

Tweets (*Sea Blue Twitter Nodes*)

Individual tweets extracted from Twitter discussions form the foundation of the knowledge graph, serving as essential data points. (For example, from Fig. 8, the node individual tweet extracted has a unique ID number 263)

Types of Tweets (*Orange Nodes Connected with Gray*)

Categorization of tweets into "Replies" and "Original" is represented by orange nodes connected with gray edges. This distinction aids in understanding the dynamics of user engagement. (From Fig. 8, this is depicted by showing Comments – 0, i.e., this tweet doesn't have any replies associated with it)

Sentiments (*Nodes in Pink*)

Nodes in pink signify sentiments attached to tweets, including "Complaint," "Appreciation," and "Neutral." These sentiment nodes provide a nuanced view of user reactions. (From Fig. 7, there are 3 major distinguishable clusters that represent the different sentiments)

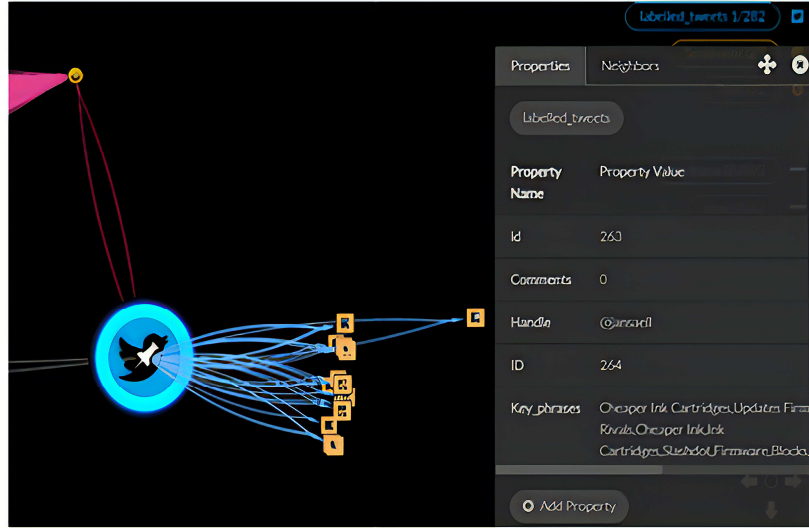


Fig. 7. Queryable Node extracted from the Graph

Key Phrases (Connect to Tweets in Light Blue)

Light blue edges connect key phrases to their respective tweets, enriching the knowledge graph with insights derived from sentiment analysis and key phrase generation. (This is depicted in Fig. 8, where tweet 263 has key phrases “Cheaper Ink Cartridges, Rivals, Firmware, Blocks, Ink”)

The generated knowledge graph serves as a valuable resource for extracting actionable insights. GraphXR, equipped with the CYPHER query language, facilitates dynamic querying and analysis. Queries can include determining product profitability, devising product development strategies, and understanding overall sentiment trends.

6 Conclusion

We have focused on analyzing customer reviews and used sentiment analysis to identify user communities and ensure that product development aligns with the customer’s needs. We have also compared our model with other deployed sentiment analysis models and scored 23% better in the F1 score metric. The generated knowledge graph is used to uncover meaningful insights and patterns from sentiment analysis and help in predicting customer behavior and market trends. To gain a better understanding of the nodes in the knowledge graph we have implemented key phrase generation for extracting high-quality key phrases, which enables us to efficiently understand the text in the nodes.

7 Future Work

In the near future, advancements in machine learning models offer opportunities to enhance the efficiency and accuracy of sentiment analysis. Integrating state-of-the-art algorithms could lead to a deeper understanding of customer sentiments. Additionally, refining and expanding knowledge graphs could involve exploring advanced graph algorithms and leveraging emerging technologies for more comprehensive representations. Building upon the knowledge graph, integrating a recommendation system could personalize user suggestions based on preferences and community trends.

References

1. Twitter website: <https://twitter.com/> last accessed 3/1/23.
2. Liu, Z., Xiao, L., Chen, J., Yu, H., & Ye, Y (2022) An Emotion-fused Medical Knowledge Graph and its Application in Decision Support. 46th International Computer Software and Applications Conference. DOI: 10.1109/COMPSAC54236.2022.00218.
3. Lotfi, C., Srinivasan, S., Ertz, M., Latrous, I., (2021). "Web Scraping Techniques and Applications: A Literature Review." scrs conference proceedings on intelligent systems, 381-394. DOI: 10.52458/978-93-91842-08-6-38.
4. Alsaity, A., & Orji, R. (2022). "Machine learning techniques for emotion detection and sentiment analysis: current state, challenges, and future directions." *Information Processing & Management*, 139-164. <https://doi.org/10.1080/0144929X.2022.2156387>
5. Tang, D., Qin, B., Feng, X., et al, (2016) "Effective LSTMs for Target-Dependent Sentiment Classification." *Proceedings of the Conference on Empirical Methods in Natural Language Processing*. DOI: <https://doi.org/10.48550/arXiv.1512.01100>
6. Papagiannopoulou, E., & Tsoumakas, G. (2019). "A review of keyphrase extraction." *WIREs Data Mining and Knowledge Discovery*. DOI: <https://doi.org/10.1002/widm.1339>
7. Peng, Y., He, L., Zhang, X., et al. (2023) "Knowledge Graphs: Opportunities and Challenges." *IEEE Transactions on Knowledge and Data Engineering*. DOI: <https://doi.org/10.1007/s10462-023-10465-9>
8. Dörpinghaus, J., Klante, S., Christian, M., Meigen, C., & Düing, C. (2022). "From social networks to knowledge graphs: A plea for interdisciplinary approaches." *Social Sciences & Humanities Open*. DOI: <https://doi.org/10.1016/j.ssaho.2022.100337>
9. Anoopkumar, M., & Rahman, A. M. J. M. Z. (2016). "A Review on Data Mining techniques and factors used in Educational Data Mining to predict student amelioration." 2016 International Conference on Data Mining and Advanced Computing (SAPIENCE), Ernakulam, India, 122-133. DOI: <https://doi.org/10.1109/SAPIENCE.2016.7684113>
10. Dinari, H. (2017). "A Survey on Graph Queries Processing: Techniques and Methods." *I. J. Computer Network and Information Security*, 48-56. DOI: <https://doi.org/10.5815/ijcnis.2017.04.06>
11. Prud'homme, E., & Seaborne, A., (2008). "Exploring SPARQL: A Query Language for RDF." In: *Proceedings of the International Semantic Web Conference*. <https://www.w3.org/TR/rdf-sparql-query/>
12. Fan, L., et al. (2023) "Multimodal knowledge graph construction of Chinese traditional operas and sentiment and genre recognition." *Journal of Multimodal User Interfaces*, 10(4), 431-449. DOI: <https://doi.org/10.1016/j.culher.2023.05.003>

13. Schoene, A., & Dethlefs, N. (2022) "Generating a linguistically inspired Knowledge Graph for fine-grained emotion classification." *International Journal of Computational Linguistics and Applications*, 15(2), 89–107. <https://aclanthology.org/2022.lrec-1.679>
14. Vizcarra, P., et al. (2020) "Knowledge-Based Sentiment Analysis and Visualization on Social Networks." *Journal of Social Media Analytics*, 8(1), 56–72. DOI: <https://doi.org/10.1007/s00354-020-00103-1>
15. Lovera, M., et al. (2021) "Sentiment Analysis in Twitter Based on Knowledge Graph and Deep Learning Classification." *IEEE Transactions on Big Data*, 9(4), 512–528. DOI: <https://doi.org/10.3390/electronics10222739>
16. Pleskach, V., Bulgakova, O., Zosimov, V., Vashchilina, E., Tumasoniene, I., (2023) "An E-Commerce Recommendation Systems Based on Analysis of Consumer Behavior Models." *International Scientific Symposium Intelligent Solutions*. https://ceur-ws.org/Vol-3538/Paper_19.pdf
17. Danyang, L., Ting, B., Jianxun, L., et al (2019). " News Graph: An Enhanced Knowledge Graph for News Recommendation "In *Proceedings of KaRS Second Workshop on Knowledge-Aware and Conversational Recommender Systems*. <https://www.microsoft.com/en-us/research/uploads/prod/2019/09/kars2019.pdf>
18. Selenium documentation. Retrieved from <https://www.selenium.dev/documentation/en/>.
19. Maas, A., Daly, R. E., Pham, P. T., Huang, D., Ng, A. Y., & Potts, C. (2011, June). "Learning Word Vectors for Sentiment Analysis." *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, 142–150 . <https://aclanthology.org/P11-1015.pdf>
20. Zhang, Y., & Wallace, (2015) B. A Sensitivity Analysis of Convolutional Neural Networks for Sentence Classification. DOI :<https://doi.org/10.48550/arXiv.1510.03820>
21. Wilson, T., Wiebe, J., & Hoffmann, P. (2005)"Recognizing Contextual Polarity in Phrase-Level Sentiment Analysis." *Proceedings of the Conference on Human Language Technology and Empirical Methods in Natural Language Processing*. <https://aclanthology.org/H05-1044.pdf>
22. Campos, R., Mangaravite, V., Pasquali, A., Jatowt, A., & Jorge, (2020) A. M. "YAKE! Keyword Extraction from Single Documents using Multiple Local Features." *Information Sciences*. DOI: <https://doi.org/10.1016/j.ins.2019.09.013>.
23. The Neo4j Team. "Neo4j: The Graph Platform." Neo4j, Inc. (2020).

***Disclosure of Interests:** The authors have no competing interests to declare that are relevant to the content on the article.